

Contact

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(LinkedIn)

Top Skills

Agile Methodologies

Software Engineering

Software Development

Certifications

Neural Networks and Deep Learning

Machine Learning : Regression and Classification

Machine Learning: Advanced Learning Algorithms

Unsupervised Learning, Recommenders, Reinforcement Learning

Publications

PhD Thesis & All Publications

Erol Akarsu

Software and Web Lifecycle Development |Quality Maintenance & Troubleshooting|DevOps Management|Big-Data Processing and Architecture|Cross-Functional Collaboration| User Research |Agile Software Development and Delivery

Richmond, Virginia, United States

Summary

Hello! I am EROL AKARSU; a tech-savvy and diligent software engineer with a Ph.D. in Computer Science and solid experience overseeing a wide range of software, application, and website development activities while aligning with web design and user experience requirements.

I specialize in developing advanced language and AI models designed to enhance home improvement projects. My expertise includes creating LLMs that offer tailored assistance to clients, ensuring their project visions are fully realized. Additionally, I innovate in AI-driven solutions for the efficient selection of client-uploaded images, guaranteeing only approved content is utilized. My goal is to streamline project workflows and elevate the quality of home improvement experiences through cutting-edge

As an innovative individual, I have extensive experience in designing and provisioning search infrastructures on Kubernetes and VM instances using Google cloud and AWS. I am recognized for designing database schemes to improve the quality of information while determining the structure and design of web pages using HTML, CSS, JavaScript, and React while resolving technical bugs. Additionally, I have strategic ability to think 'outside the box and resolve technical issues for operational efficiency.

My professional experience includes overseeing team projects, designing software, analyzing data and solving complex problems on a daily basis to ensure a high level of quality for the software I help to create. I was recognized for enhancing cloud-based search experience while near-real-time indexing items and developing processes for ingesting products into Couchbase and indexing them on Solr.

My forte includes installing architecture using Docker swarm in liaison with infrastructure team, software engineers, and testing team. I am highly capable of configuring and running distributed crawling (Apache Nutch) on 15K+ sites and building a distributed ½ TB indexed database (Apache SOLR) through Apache Hadoop.

★ Acted as main architect to move search infrastructure to Cloud environment.

★ Built data pipelines for search data with Apache beam and spark language while working as Data Scientist/Engineer.

★ Created Microservices in Spring Boot, Ruby on Rails, and Node.js and JBoss.

Connect with me today to find out how I'll make your mission my mission, to help bring ALL of your business objectives into focus! Please feel free to contact me at eakarsu@gmail.com with any thoughts, comments, or questions about my work. I'm always interested in making new professional acquaintances.

Experience

Lowe's Companies, Inc.

5 years 6 months

Lead Machine Learning Architect

January 2024 - Present (2 years 3 months)

Charlotte, North Carolina, United States

We are developing an advanced system to curate and select the most optimal images from a vast image database using state-of-the-art Machine Learning (ML) transformer models. The selected images will be prominently displayed as product images, enhancing visual appeal and user engagement on our platform.

In addition to our image selection initiative, we are undertaking a Project Quota Assistance system powered by Generative Artificial Intelligence (AI). This project is designed to help users promptly ask questions about quota and receive immediate, accurate answers, thereby optimizing resource allocation and enhancing project management efficiency.

Principal Software Engineer

October 2020 - Present (5 years 6 months)

Mooresville, North Carolina, United States

Design and build various features for Lowe's.com such as workflow applications for Apache Beam data pipelines and Elastic Search frameworks. Leverage technical product skills and people-leadership abilities to create org-wide impact and lead broad project initiatives. Involved in project planning efforts such as contracting, external contracting and budget management.

- Engineered the creation of pivotal features on Lowe's.com such as free delivery, expedited delivery and promotions by leading end-to-end software lifecycle and developing Spark applications.
- Significantly reduced client cost while elevating feature quality by planning and deploying automated build processes to optimize long-term maintenance costs of the solution.
- Spearheaded project roadmap, implementation process and launch plan for Spark application development process by identifying business need and serving as single-threaded leader between senior VPs and various engineering and design teams. Worked to define path forward by prioritizing end-user impact while maintaining regulatory compliance.
- Enhanced cloud-based search experience while near-real-time indexing items and developing processes for ingesting products into Couch Base and indexing them on Solr.

Wipro Limited

Senior Software Architect

May 2018 - October 2020 (2 years 6 months)

Richmond, Virginia Area

Architected various products on Solr geared towards blockchain and cryptocurrency customers. Created ground-breaking features such as profit sharing with nominal interest fees, cutting-edge financing for Square lending, and security schema for borrowed tokens across various cryptocurrency projects. Utilized knowledge of Solr, Couchbase and Spark Scala to update product data, maintain data indices and industrialize Spark Scala's 20+ step indexing pipeline.

- Utilized cloud-based tools and APIs, such as Terraform for cloud resource creation, Packer for image creation, Ansible resource provisioning, and Jenkins for auto-deployment while constructing Lowe's enterprise search infrastructure on the Google cloud.
- Established cloud-based load balancers, firewalls, and clusters while developing dozens of Jenkins jobs to proceed with Terraform scripts and design images with Packer.

- Alleviated product indexing time by 9 times (from 360 to 40 minutes for 5 million products) while creating an ETL process in Scala to transform product data in Couchbase with Spark, Couchbase, and Solr.
- Enhanced operational efficiency while understanding blockchain-related tools and technologies, such as NextJS Web Frontend, Web3, Remix, and Solidity.

pExchange

Senior Software Developer

December 2017 - March 2018 (4 months)

Richmond, Virginia Area

Augmented productivity by assessing and suggesting a standard software toolset for team members. Ensured effective digital media workflow while designing and recommending software automation solutions.

- Utilized Spring boot, Mongo, Angular, and Apache Shiro for user roles and permissions while designing a user management system to prepare presidential budgets for specific programs.

Capital One

Master Software Engineer

August 2016 - December 2017 (1 year 5 months)

Richmond, Virginia Area

Led efforts to design data flows and sandbox environments for API development and service virtualization frameworks. Orchestrated the development of novel features such as general simulators and Spring/Spring Boot frameworks while collaborating with end-users and increasing their understanding of API networks.

- Enhanced multiple web features and presented them within a company-wide meeting. Worked to truly automate website data flow by defining inputs and outputs for various microservices, designing data flows and graphs to project key outputs
- Led a team to overhaul the design of developer.capitalone.com website and regression suite while utilizing Protractor, Cucumber, Spring Framework, NodeJS, JavaScript, Express, AngularJS, and Bower/Gulp site.
- Produced Cucumber test feature specs and fully automated API testing without a user input via MVC, Swagger, FSM, Nodejs/JavaScript, and Graph Computing.

Self Employed

Independent Software Consultant

September 2003 - August 2017 (14 years)

Oversaw a large number of independent projects for high-priority customers such as the US Government by employing in-depth knowledge of industry-relevant web technologies and client product management skills.

- Renovated government website with Angular and NodeJS. Worked with client product managers, engineers to gather requirements, build the design, end to end delivery leading the entire software development life cycle.
- Built the entire end-to-end Turkish search engine for a food retail chain by building big data pipelines, enabling product searching, and utilizing tokenizers and Apache to filter relevant results via Nutch/SOLR/Hadoop/Node.js/Bootstrap. Utilized SOLR engine and Node.js to develop a specialized search engine with 15 ranking criteria.
- Enabled users in designing composite web services with continuous access to composite service while applying for authentication and authorization of requests (username/password, SSL certificates, and token authentication). Developed and implemented SOA Router/Proxy via NET technologies, including C#, Microsoft LINQ, WCF, MS-SQL 2005, MS-IIS Server, and Oracle 10g to secure intra-company web services. Supervised OpenDNS-based power grid stimulation through a Python-developed tool.
- Organized technical environment to call Java Web services from PL/SQL while liaising with on design team for the new Navy portal with several Oracle portal products (WCI, Web Center) and supervising portals on Ubuntu Linux, UNIX, and Windows.
- Created and regulated Oracle WCI API for the Navy while building and executing automated portal.
- Ensured 99.5% extraction success rate by generating and implementing the rule extraction in XQUERY functional language to 700K lines of COBOL code (237 files).

Ferguson Enterprises

SENIOR SOFTWARE ANALYST/TECHNICAL LEAD

July 2007 - July 2016 (9 years 1 month)

Worked on several web development projects

Cisco Systems

Software Engineer

November 1999 - July 2007 (7 years 9 months)

Worked on unified communications software

Education

Syracuse University

PhD, IDE for Grid Computing · (1994 - 1999)

Syracuse University

Master of Science (M.S.), Computer Science · (1994 - 1996)

Istanbul Technical University

Master of Science (M.S.), Computer Engineering · (1991 - 1993)

Ege University

Bachelor of Science (B.S.), Computer Engineering · (1986 - 1991)